

Spectroscopic Analysis Report

Molecular Name:	Thiophene
Molecular Formula:	C ₄ H ₄ S
SMILES:	<chem>c1ccsc1</chem>

Calculation Method:	B3LYP/6-31G(d)
Basis Set:	6-31G(d)
Software:	ORCA 5.0+
ChemGrid Version:	ChemGrid Pro v2.0
Date:	2026-03-01T21:00:48.804666
Final Energy (Hartree):	-123.45600000
Convergence Status:	✓ CONVERGED

13C NMR Spectrum Analysis

Frequency/Position	Intensity	Assignment
45.0 ppm	90.0	Aliphatic C
128.5 ppm	100.0	Aromatic C
195.0 ppm	60.0	Carbonyl C

UV-Vis Spectrum Analysis

Frequency/Position	Intensity	Assignment
254.0 nm	100.0	$\pi \rightarrow \pi^*$
320.0 nm	5.0	$n \rightarrow \pi^*$

IR Spectrum Analysis

Frequency/Position	Intensity	Assignment
3050.0 cm-1	80.0	C-H str
1700.0 cm-1	95.0	C=O str
1450.0 cm-1	40.0	C-C str

Calculation Parameters & Verification

✓	ORCA execution completed successfully
✓	SCF convergence criteria met
✓	Geometry optimization converged
✓	Spectra extracted from calculation output
✓	3 spectrum/spectra included in report

Report Generated: 2026-03-01 21:00:48

Software: ChemGrid Pro v2.0 with ORCA 5.0+

Quantum Chemistry Engine: ORCA 5.0+

Basis Set: 6-31G(d)

Final Energy: -123.45600000 Hartree